

BUCHS-AARAU, Switzerland

WMO: 066330

Lat: 47.3844N

Lon: 8.0794E

Elev: 389

StdP: 96.74

Time Zone: 1.00 (EUC)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-8.3	-6.0	-12.0	1.4	-6.9	-9.3	1.8	-4.5	8.7	8.0	7.4	6.8	1.8	50	0.459

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.4	31.4	20.1	29.3	19.5	27.4	18.8	21.1	28.7	20.2	27.3	19.5	25.9	2.0	40

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.7	14.2	23.2	18.0	13.5	22.2	17.2	12.9	21.5	62.6	28.9	59.7	27.5	57.1	25.9	25.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 6.7	5.3	4.3	-11.2	34.2	2.8	1.6	-13.2	35.3	-14.9	36.3	-16.4	37.2	-18.5	38.4		
(5)			-11.6	22.8	2.9	1.1	-13.8	23.6	-15.5	24.2	-17.1	24.8	-19.2	25.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	10.5	1.3	2.0	6.2	10.5	14.2	18.4	20.0	19.3	15.2	10.9	5.5	1.9		
	DBStd	7.59	3.95	4.06	3.90	3.67	3.54	3.63	3.06	3.20	3.12	3.64	3.42	3.62		
	HDD10.0	1089	269	223	127	38	6	1	0	0	2	34	140	251		
	HDD18.3	3044	527	457	377	234	135	44	18	27	102	229	386	509		
	CDD10.0	1273	1	0	8	54	136	252	309	288	158	63	4	0		
	CDD18.3	186	0	0	0	0	7	46	69	57	8	0	0	0		
	CDH23.3	1801	0	0	0	14	93	449	667	504	72	1	0	0		
	CDH26.7	576	0	0	0	1	16	143	232	177	7	0	0	0		
(14) Wind	WSAvg	1.5	1.7	1.7	1.8	1.5	1.5	1.4	1.3	1.1	1.1	1.1	1.4	1.7		
(15) Precipitation	PrecAvg	1098	74	71	72	82	110	112	117	111	83	85	89	91		
	PrecMax	1408	186	186	247	174	204	207	206	211	151	156	197	195		
	PrecMin	748	15	22	25	8	30	36	36	18	41	10	4	0		
	PrecStd	162	47	41	49	44	44	42	42	45	34	39	49	47		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.1	14.8	19.7	25.3	28.8	33.1	34.1	34.2	27.6	22.1	16.4	11.6		
		MCWB	9.2	8.7	11.2	14.4	18.2	20.8	20.7	20.4	18.7	15.7	11.8	9.5		
	2%	DB	10.4	11.7	17.6	22.7	25.7	30.3	31.4	30.8	25.6	19.7	13.6	9.9		
		MCWB	7.9	7.2	10.0	13.1	16.7	19.8	19.8	20.2	18.1	14.9	10.8	8.3		
	5%	DB	8.6	9.6	15.4	20.6	23.7	28.2	29.2	28.4	23.2	17.9	11.8	8.6		
		MCWB	6.6	6.3	9.2	12.4	15.4	19.2	19.3	19.6	16.9	14.1	9.7	7.0		
	10%	DB	6.9	7.7	12.9	18.2	21.3	25.9	27.2	26.1	21.0	16.1	10.2	7.1		
		MCWB	5.3	5.1	8.2	11.2	14.4	18.2	18.6	18.9	16.1	13.4	8.6	5.8		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.0	9.4	12.0	15.1	18.8	22.5	21.9	22.0	19.9	16.7	12.4	9.8		
		MCDB	11.5	13.2	18.0	23.2	26.1	30.2	30.8	29.5	25.9	20.4	15.3	11.2		
	2%	WB	8.2	7.8	10.7	13.9	17.6	20.9	20.7	20.9	18.7	15.6	11.1	8.5		
		MCDB	10.0	11.0	15.7	21.3	24.1	28.1	28.6	28.4	23.8	18.5	13.3	9.8		
	5%	WB	6.7	6.6	9.7	12.8	16.4	19.8	20.0	20.1	17.6	14.6	9.9	7.2		
		MCDB	8.5	9.5	14.0	19.1	21.7	26.4	27.1	27.1	22.0	17.2	11.7	8.4		
	10%	WB	5.4	5.3	8.7	11.7	15.2	18.7	19.2	19.2	16.6	13.6	8.8	5.8		
		MCDB	6.9	7.5	12.6	17.3	20.0	24.7	25.6	25.0	20.2	15.7	10.0	7.0		
(35) Mean Daily Temperature Range	5% DB	MDBR	4.9	6.9	9.5	10.8	10.2	10.9	11.4	10.8	9.9	7.9	5.6	4.4		
		MCDBR	7.2	11.1	14.4	15.8	14.8	15.2	15.7	15.1	13.4	10.4	8.3	5.9		
	5% WB	MCWBR	5.1	6.9	7.9	7.5	6.6	6.0	5.7	5.8	6.5	5.8	5.6	4.4		
		MCWBR	6.6	10.0	12.6	14.2	12.8	13.8	13.8	13.6	11.9	9.5	7.8	5.2		
(40) Clear-Sky Solar Irradiance	taub	taub	0.300	0.327	0.363	0.395	0.407	0.411	0.412	0.404	0.381	0.358	0.324	0.297		
		taud	2.438	2.364	2.311	2.257	2.267	2.299	2.289	2.326	2.374	2.418	2.461	2.446		
	Ebn at Noon	Ebn at Noon	787	834	853	857	858	854	847	838	825	786	749	744		
		Edh at Noon	67	88	109	126	130	127	126	117	102	84	66	59		
(44) All-Sky Solar Radiation	RadAvg	RadAvg	1.16	1.99	3.21	4.34	5.00	5.82	5.61	4.89	3.69	2.32	1.28	0.94		
	RadStd	RadStd	0.14	0.29	0.46	0.64	0.50	0.54	0.51	0.48	0.37	0.24	0.16	0.16		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	+0.37	N/A	N/A	N/A	N/A		
(47) Regional (117 neighbors)	N/A	N/A	N/A	+0.73	+0.43	+0.34	N/A	N/A	+56	+14	

Nomenclature: See separate page